

KHORINSK, Russian FederationWMO: **307390**Lat: **52.167N**Lon: **109.785E**Elev: **2185**StdP: **13.57**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.4	-34.0	-42.0	0.5	-37.8	-37.9	0.7	-33.9	18.7	5.5	17.4	4.0	0.4	50	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	86.2	65.1	82.9	63.7	79.6	62.3	67.7	81.1	65.9	77.9	64.2	75.2	6.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	95.1	71.9	61.8	89.5	70.4	60.0	83.7	69.8	33.5	81.4	32.0	77.9	30.6	75.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.5	16.4	14.3	DB	-42.0	92.6	6.4	5.1	-46.6	96.3	-50.4	99.3	-54.0	102.1	-58.7	105.8
(5)				WB	-41.2	71.2	5.4	3.0	-45.1	73.3	-48.2	75.1	-51.3	76.8	-55.2	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	-12.9	-3.8	16.2	35.1	48.3	62.7	67.8	62.1	47.9	30.4	10.3	-6.6	(6)
(7)		DBStd	29.19	12.70	11.62	11.73	8.02	7.69	7.24	5.64	6.19	7.06	8.60	11.94	12.88	(7)
(8)		HDD50	8760	1950	1506	1047	449	125	4	0	2	123	608	1192	1754	(8)
(9)		HDD65	13005	2415	1926	1512	897	517	126	35	130	514	1072	1642	2219	(9)
(10)		CDD50	1452	0	0	0	2	74	386	552	377	60	1	0	0	(10)
(11)		CDD65	220	0	0	0	0	1	57	122	40	0	0	0	0	(11)
(12)		CDH74	2249	0	0	0	0	51	671	1156	362	9	0	0	0	(12)
(13)		CDH80	621	0	0	0	0	5	179	374	63	0	0	0	0	(13)
(14)	Wind	WSAvg	5.4	4.1	3.9	5.4	6.9	7.2	6.4	5.4	5.1	5.6	5.3	5.4	4.3	(14)
(15)	Precipitation	PrecAvg	11.3	0.1	0.1	0.1	0.4	0.7	1.8	3.1	2.6	1.5	0.4	0.2	0.2	(15)
(16)		PrecMax	16.3	0.2	0.4	0.3	1.0	2.1	6.0	8.3	5.9	3.1	1.5	0.6	1.2	(16)
(17)		PrecMin	6.4	0.0	0.0	0.0	0.0	0.1	0.4	0.3	0.7	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	3.0	0.1	0.1	0.1	0.3	0.5	1.4	1.7	1.3	0.8	0.4	0.1	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	28.7	47.7	65.0	80.7	89.6	92.1	85.6	74.8	58.4	40.6	25.5	(19)
(20)		0.4%	MCWB	19.3	24.3	37.4	48.1	58.6	67.0	68.2	65.5	57.8	46.6	33.2	22.5	(20)
(21)		2%	DB	16.5	23.4	42.3	59.0	74.9	85.0	87.7	81.3	69.1	51.2	35.8	21.7	(21)
(22)		2%	MCWB	14.3	20.0	33.1	44.1	54.7	63.3	66.8	63.6	55.6	41.8	30.6	19.2	(22)
(23)		5%	DB	10.7	18.8	38.3	54.2	69.8	81.5	84.2	78.2	64.7	46.9	31.4	17.4	(23)
(24)		5%	MCWB	8.9	16.2	30.6	41.6	52.3	61.7	65.2	62.6	53.0	39.2	27.6	15.4	(24)
(25)		10%	DB	5.7	14.2	33.8	49.8	64.8	77.9	80.8	74.6	60.5	42.8	27.6	12.8	(25)
(26)		10%	MCWB	4.3	12.1	27.8	38.6	49.0	60.4	63.8	60.9	50.7	36.7	24.4	11.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	24.6	37.5	48.5	59.3	68.9	71.7	68.3	60.4	47.5	33.9	22.7	(27)
(28)		0.4%	MCDWB	21.6	28.4	47.3	63.8	76.9	86.8	85.8	79.6	70.6	56.5	39.4	25.2	(28)
(29)		2%	WB	14.5	20.3	33.6	44.6	55.9	65.7	68.8	65.8	56.8	42.9	30.5	19.1	(29)
(30)		2%	MCDWB	16.3	23.4	41.4	57.4	73.3	79.9	83.1	75.9	66.9	50.1	35.4	21.6	(30)
(31)		5%	WB	8.9	16.3	30.8	42.0	52.9	63.7	67.0	64.0	54.3	39.7	27.8	15.5	(31)
(32)		5%	MCDWB	10.5	18.9	38.0	53.1	67.6	76.8	79.8	74.0	63.4	46.1	31.4	17.4	(32)
(33)		10%	WB	4.3	12.1	28.0	39.3	50.2	61.7	65.3	62.4	51.4	36.9	24.5	11.4	(33)
(34)		10%	MCDWB	5.6	14.2	33.6	49.1	63.5	74.2	76.9	72.2	59.6	42.4	27.5	12.9	(34)
(35)	Mean Daily Temperature Range		MDBR	17.2	23.6	23.9	20.7	21.9	21.5	19.7	18.9	19.3	18.7	17.8	16.2	(35)
(36)		5% DB	MCDBR	16.5	25.0	26.3	28.2	31.7	28.7	26.0	25.2	25.8	25.0	19.6	15.6	(36)
(37)		5% DB	MCWBR	15.2	22.1	18.4	17.6	17.5	12.9	10.3	11.2	14.1	16.1	15.2	13.9	(37)
(38)		5% WB	MCDBR	16.5	24.7	25.5	26.3	28.7	24.2	21.9	20.1	23.6	23.7	19.5	15.7	(38)
(39)		5% WB	MCWBR	15.3	21.8	17.9	17.1	16.5	11.7	10.0	9.8	13.4	15.6	15.4	14.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.281	0.313	0.363	0.450	0.445	0.412	0.374	0.332	0.300	0.279	0.259	(40)	
(41)		taud	2.201	2.186	2.219	2.243	2.013	2.151	2.309	2.395	2.456	2.458	2.301	2.175	(41)	
(42)		Ebn at Noon	236	264	278	276	254	257	265	270	271	259	226	210	(42)	
(43)		Edh at Noon	21	29	35	39	52	46	38	33	28	23	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	406	739	1203	1600	1815	1903	1759	1489	1138	719	407	295	(44)	
(45)		RadStd	22	30	38	103	107	122	137	77	94	60	30	19	(45)	